

Cloud Computing TASK – 4

- 1. Create a comparison table showing which security responsibilities belong to the cloud provider and which belong to the user for a food delivery app like Zomato running on AWS, Azure, or Google Cloud.**

Ans :

Security Area	Cloud Provider Responsibility	User (Food Delivery App) Responsibility
Physical Security	Protects data centers, servers, and physical infrastructure from unauthorized access.	No direct responsibility for physical security.
Network Security	Provides secure networking infrastructure, firewalls, and protection against large-scale attacks.	Configures network rules, security groups, and access permissions for the application.
Operating System	Maintains the security of the underlying cloud infrastructure and managed services.	Updates and secures the operating system of virtual machines if using services like EC2 or virtual machines.
Application Security	Provides secure cloud platforms and tools for application development.	Secures the food delivery app code, fixes vulnerabilities,

Security Area	Cloud Provider Responsibility	User (Food Delivery App) Responsibility
		and performs security testing.
Data Protection	Provides encryption tools, backup options, and secure storage services.	Protects customer data, manages encryption keys, and controls who can access user information.
Identity and Access Management	Provides IAM services and authentication tools.	Creates user roles, manages permissions, and ensures employees have only required access.
Database Security	Secures managed database infrastructure and hardware.	Protects customer records, configures database access, and manages sensitive information securely.

2. Sign up for a free trial on AWS, Azure, or Google Cloud, and create a new user account with limited permissions (for example, only allow access to storage buckets or databases, but not to billing or admin settings).

Ans :

- Logged in to the AWS Management Console using my AWS account.

- Opened the IAM (Identity and Access Management) service from the search bar.
- Selected Users and clicked Create User.
- Entered a username (for example, storage-user).
- Enabled access for the user and created the user account.
- Created a permission policy with limited access, such as Amazon S3 Read Only Access.
- Attached the policy to the new user so the user can access storage buckets but cannot access billing or administrative settings.
- Reviewed the permissions and completed user creation.
- Tested the new user login to confirm that only allowed services were accessible.

**3. Configure a custom po to upload images to a storage bucket but denies them permission to delete any files.

Hint: Look for storage service permissions such as 'PutObject' and 'DeleteObject' in the policy editor.**

Ans :

- Logged in to the AWS Management Console.
- Opened the IAM (Identity and Access Management) service.
- Selected Policies and clicked Create Policy.
- Chose S3 as the service.
- Added permission to upload images by allowing the action:
 - s3:PutObject
- Restricted deletion by denying the action:
 - s3:DeleteObject

- Selected the specific S3 bucket where images can be uploaded.
- Reviewed the policy and created the custom policy.
- Attached this policy to the limited-permission user created earlier.
- Tested the permissions by uploading an image and verifying that deleting files was not allowed.

**4. Simulate a real-world scenario: Imagine you are part of a team managing a cricket ticket booking app (like BookMyShow). Assign the following roles using your cloud provider's IAM or equivalent: one user can only view booking data, another can edit booking data, and a third can manage user accounts.

Hint: Use built-in roles or create custom ones if needed.**

Ans :

- Create IAM Users

Create three users:

User	Role	Responsibility
booking-viewer	Booking Data Viewer	Can only view ticket bookings
booking-editor	Booking Data Editor	Can view and modify booking details
user-admin	User Account Manager	Can create, update, and delete user accounts

- Create IAM Policies

Role 1: Booking Data Viewer

Purpose: Customer support employee who only needs to check booking details.

Permissions:

- Read booking records
- View ticket status
- View customer booking history

Role 2: Booking Data Editor

Purpose: Booking operations team that manages ticket reservations.

Permissions:

- View bookings
- Update booking status
- Cancel or modify reservations

Role 3: User Account Manager

Purpose: Application administrator responsible for customer accounts.

Permissions:

- Create new users
- Update user details
- Remove inactive accounts

Final IAM Structure

Cricket Ticket Booking Application

IAM

booking-viewer	booking-editor	user-admin
Read Only	Read + Edit	Manage Users
Bookings	Bookings	Accounts

5. Find and describe one real security incident from the last 3 years where a cloud misconfiguration (wrong permissions or policies) exposed user data. Explain in 3-4 lines how applying the Shared Responsibility Model could have prevented it.

Ans :

- In 2023, a misconfigured cloud email server hosted on Microsoft Azure exposed sensitive email data to the public internet without requiring authentication. The exposure affected around 20,000 individuals, whose information was later subject to breach notification. The issue was caused by incorrect cloud configuration settings rather than a failure of the cloud provider itself.
- How the Shared Responsibility Model could have prevented it.
 - Under the Shared Responsibility Model, the cloud provider is responsible for securing the cloud infrastructure, while the customer is responsible for configuring services, access controls, and permissions correctly. Regular security audits, least-privilege access policies, and automated configuration checks could have detected the publicly accessible email server.

Proper monitoring and IAM reviews would have reduced the risk of unauthorized data exposure.